



From Headlines to Hedges

An AI/NLP risk engine that turns news & social text into sentiment, event and impact signals for index and credit risk

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S&P Global × Crisil Campus Hackathon 2026 · Phase III case study
Modules delivered: A (sentiment index rebalancing) and B (event-driven stress testing)

Problem & approach

The problem. Market-moving information arrives as unstructured text — a downgrade, a sanctions headline, a CEO tweet — long before it shows up in prices, spreads or ratings. Risk teams need it **quantified, classified and routed** to the processes that act on it.

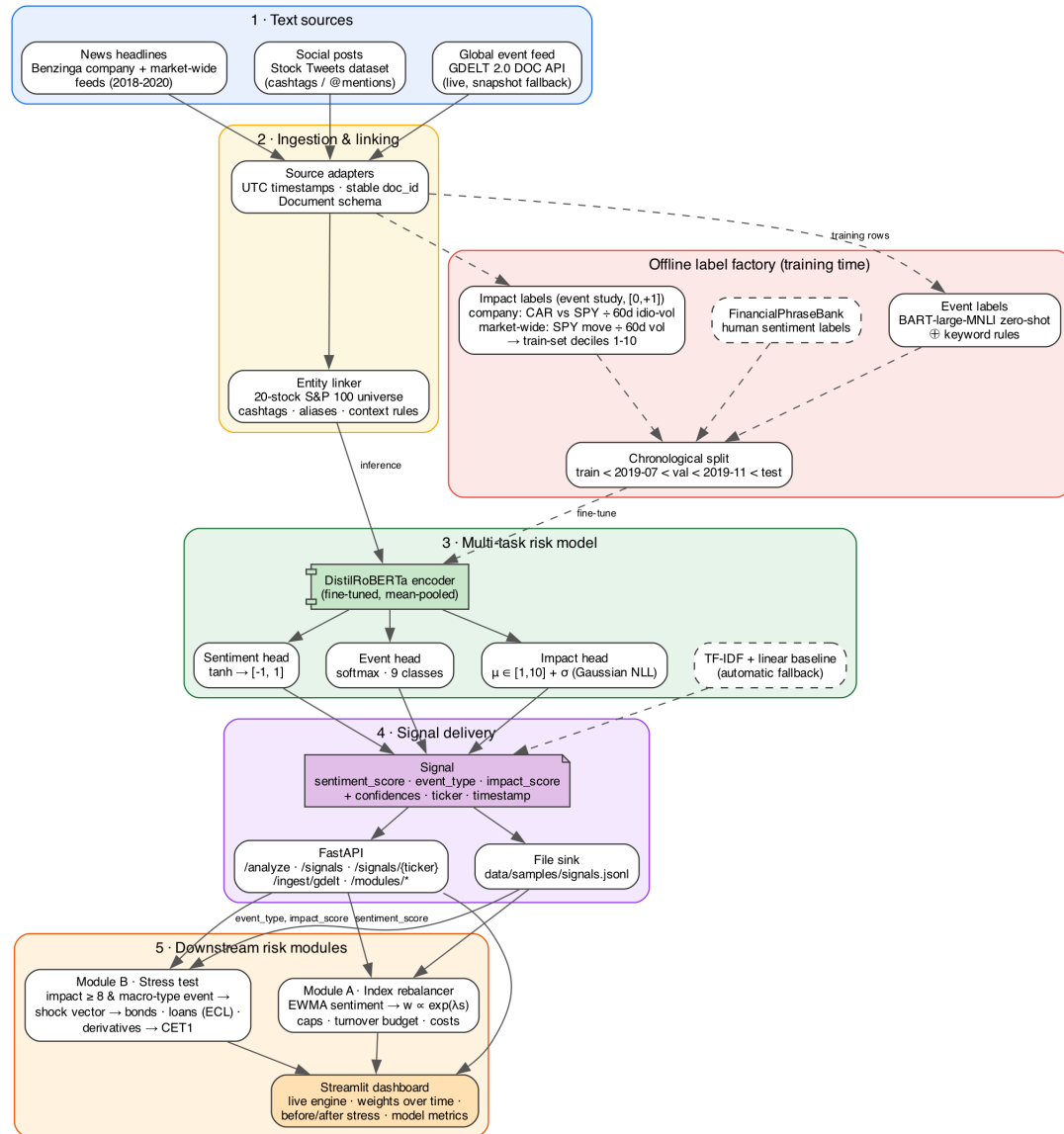
The approach.

- **Three outputs per text:** sentiment in $[-1, 1]$, one of 9 event types, and an impact score 1-10 that is *anchored to realised market reactions*, not a guess.
- **Three live-capable sources:** financial news (Benzinga), social posts (Stock Tweets), and the GDELT global event feed for macro & geopolitical news.
- **One shared model:** a fine-tuned multi-task transformer, with a TF-IDF baseline as a measured yardstick and automatic fallback.
- **Two consumers:** an index desk (Module A) and a credit-risk / capital desk (Module B), fed through a REST API and a JSONL file.

Design choices that matter to a risk user

- **No look-ahead anywhere.** Chronological train/val/test split; weights formed on day t trade on $t+1$; post-16:00 ET headlines roll to the next session.
- **Impact = event study, at the right scope.** Company news: abnormal return over $[0, +1]$ vs SPY $\div$ 60-day idiosyncratic vol. Market-wide news: the SPY move $\div$ its own 60-day vol (a market model would net systemic shocks out). Train-set deciles, so a 9 means a top-decile surprise.
- **Honest labels.** Event classes come from BART-MNLI zero-shot reconciled with keyword rules, and a 268-row hand-labelled set is the independent check.
- **Uncertainty is first-class.** The impact head predicts a mean and a variance, so every signal carries a confidence that downstream modules can threshold.
- **Regulatory framing.** Module B speaks the bank's language: IFRS 9 ECL & SICR staging, standardised RWA, CET1 vs the 7% buffer.

System design



- **Ingestion:** per-source adapters normalise to a single Document schema (UTC time, stable id, entity hints).
- **Entity linking:** cashtags, aliases and context rules map text to a 20-stock S&P 100 universe; ambiguous names (Apple, Meta) need financial context. Unlinked macro news becomes an event-level signal.
- **Inference:** one encoder pass yields all three outputs and their confidences.
- **Delivery:** FastAPI (/analyze, /signals, /signals/{ticker}, /ingest/gdelt, /modules/*) and signals.jsonl.
- **Consumers:** Module A subscribes to sentiment; Module B to event type + impact. A Streamlit dashboard exposes all of it live.

Implementation

Training data (public Kaggle sources; all files in data/)

Source	Rows	Supplies
FinancialPhraseBank	4,846	human sentiment labels
Benzinga company news	13,000	events + company impact
Benzinga market-wide news	3,655	market impact (SPY move)
Stock Tweets	3,481	social events + impact

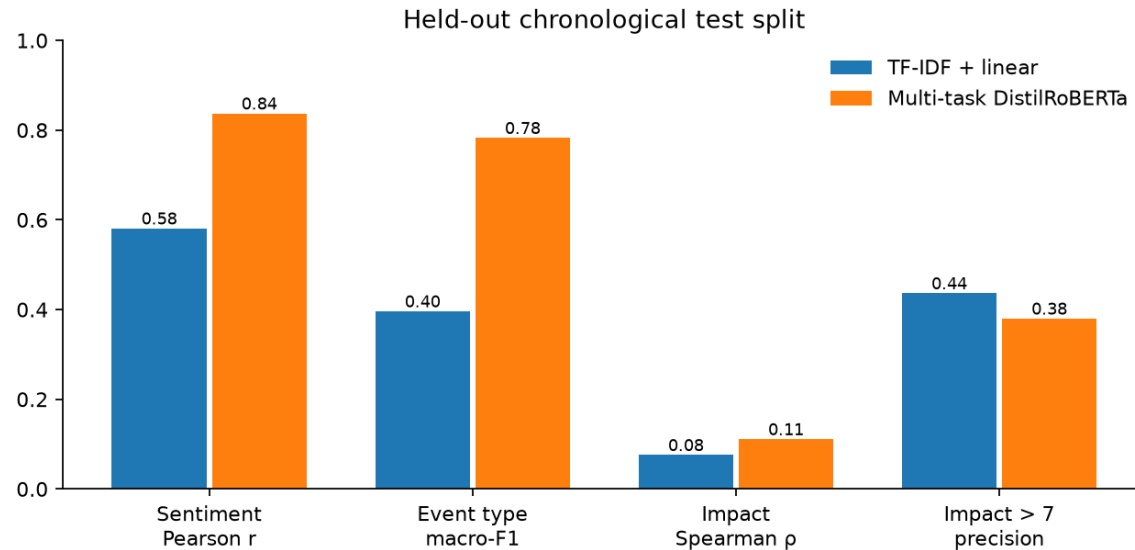
Total 24,982 rows · 4,846 sentiment · 10,554 event · 20,136 impact labels. Test split = news published after 2019-11-01 (includes the COVID-19 crash).

- **Model:** DistilRoBERTa encoder, mean pooling, three heads — tanh sentiment regression, 9-way event softmax, heteroscedastic impact (mean + log-variance, Gaussian NLL).
- **Masked multi-task loss:** each row only trains the heads it has labels for; class-balanced event weights; majority class capped at 40%.
- **Training:** AdamW, heads at 10× encoder LR, warm-up, early stopping on a composite validation score. Runs on an Apple M3 (MPS) in minutes.
- **Engineering:** 44 pytest tests (no-look-ahead, trigger rule, bond / swap signs, API), pinned requirements, make targets for every stage.

Downstream modules

- **Module A:** per-stock EWMA of impact-weighted sentiment (half-life 5d); target $w \propto (1/N) \cdot \exp(\lambda \cdot s)$, $\lambda=1$; bounds [1%, 12%]; 10% daily turnover budget; 5 bps costs. λ , half-life and weighting chosen on the in-sample window only (24-point grid).
- **Module B:** synthetic wholesale book of 144 positions — corporate loans, IG/HY & sovereign bonds, rates swaps, equity TRS & puts, FX forwards, commodity swaps, CDS.
- **Trigger:** impact ≥ 8 , a systemic event type, adverse sentiment and event confidence ≥ 0.6 ; one scenario per event-day. Shock vector (equity, rates, IG/HY spreads, USD, commodities, sector PD multipliers) is scaled by impact/10.
- **Revaluation:** bonds by duration + convexity; loans by IFRS 9 ECL with SICR staging (2× PD) and floating-rate PD uplift; derivatives by DV01 / delta-gamma / CS01; capital via standardised RWA → CET1.

Key results



0.84

sentiment Pearson r (test)

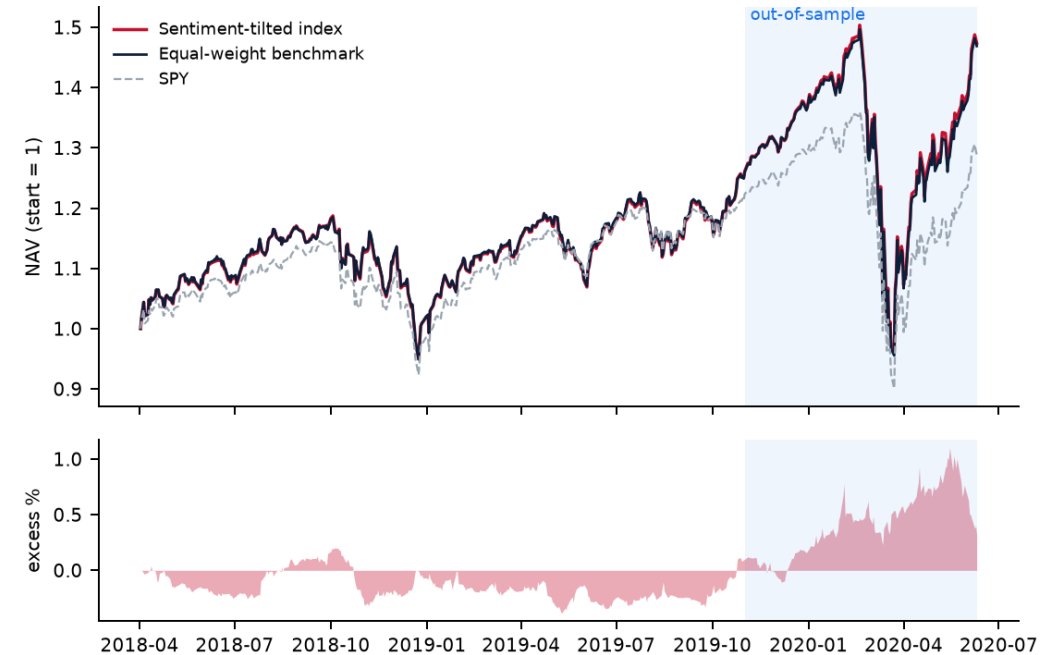
0.78

event macro-F1 (test)

0.11

impact Spearman ρ (test)

multi-task DistilRoBERTa vs TF-IDF baseline on the same held-out test split: sentiment r 0.58 → 0.84, event macro-F1 0.40 → 0.78, impact ρ 0.08 → 0.11. Latency 9 ms p50 per headline (MPS), 12 ms on CPU; 912 headlines/s batched.



+0.2%

excess return vs EW (OOS)

0.30

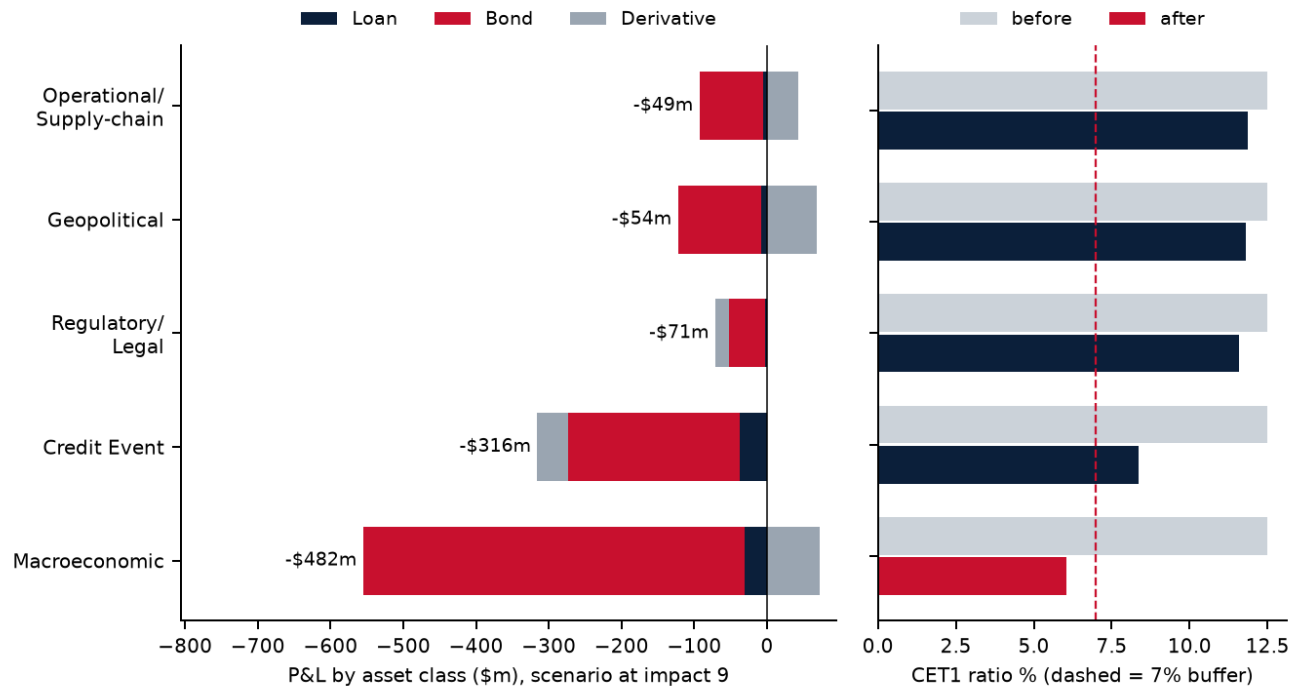
information ratio (OOS)

0.021

sentiment IC (OOS, $t=0.9$)

Module A, out-of-sample 2019-11-01 → 2020-06-10 (through the COVID crash): Sharpe 0.81 vs 0.80 for equal weight, tracking error 1.1%, daily hit rate 56%, avg turnover 1.52%/day. Signals from *multitask-distilroberta-base-v1.0.0*; 27,669 signals in the stream.

Domain impact



Reference scenarios at impact 9 on the synthetic \$9.1bn book. A Macroeconomic shock (rates +, spreads wider) costs **-\$482m** (-5.3%), moves 5 loans to Stage 2 and takes CET1 from 12.5% to **6.1%** — below the 7% buffer. A Credit Event moves 12 loans to Stage 2 (ECL \$23m → \$60m). Pay-fixed swaps and CDS protection offset part of the losses.

What this changes for a risk desk

- **Minutes, not days.** A downgrade or sanctions headline becomes a sized, typed, scored signal and — if severe — an automatic stress run with a capital read-out.
- **Early-warning for credit.** Sentiment and Credit-Event signals per obligor flag SICR candidates before ratings move; ECL and Stage-2 migration are quantified immediately.
- **Capital planning.** The CET1 waterfall shows which desks and sectors consume the buffer — the input for hedging or limit decisions.
- **Systematic overlay for index products.** Module A's tilt is bounded, turnover-capped and cost-aware; weighting sentiment by predicted impact beat plain averaging in-sample.
- **Audit trail.** Every signal keeps source, timestamp, model version and confidence; history replay scanned 27,669 signals and triggered 162 stress tests.

Limitations & next steps

- **Label quality.** Event labels are zero-shot + rules (a teacher), so test macro-F1 measures agreement with the teacher, not ground truth. The 268-row hand-labelled event set is prepared but not yet fully annotated.
- **Impact is noisy by nature.** Daily returns are dominated by non-news variance: company-impact ρ is 0.12 on test, and the day-level market-wide label does not yet generalise to the COVID window (ρ 0.00). Intraday prices would sharpen both.
- **Data window.** Free Benzinga and Stock Tweets data end in mid-2020 and cover 20 large caps; social posts are sparse outside 2018 H2. GDELT is live but rate-limited.
- **Module B is a sensitivity model.** Linear / quadratic revaluation, a static balance sheet and a hand-calibrated scenario library; no full revaluation, liquidity or second-round effects.
- **Module A is a 2-year backtest** on 20 names with simple costs; the out-of-sample edge is small (IC t-stat 0.9, not significant) and needs a longer, wider universe before any capital is allocated.

Next steps

- Active-learning loop: route low-confidence signals to analysts via the labelling tool; retrain weekly.
- Larger finance encoders (FinBERT / domain-adapted RoBERTa) and entity-level sentiment for multi-company headlines.
- Intraday event study and sector-relative abnormal returns for sharper impact labels.
- Obligor-level signal aggregation feeding PD overlays; reverse stress testing for the CET1 buffer.
- Production hardening: streaming ingestion (Kafka), model registry, drift monitoring on sentiment and class mix.

AI-assistance disclosure: code was written with an AI coding assistant under the author's direction; all design decisions, data choices and results were reviewed by the author.